UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

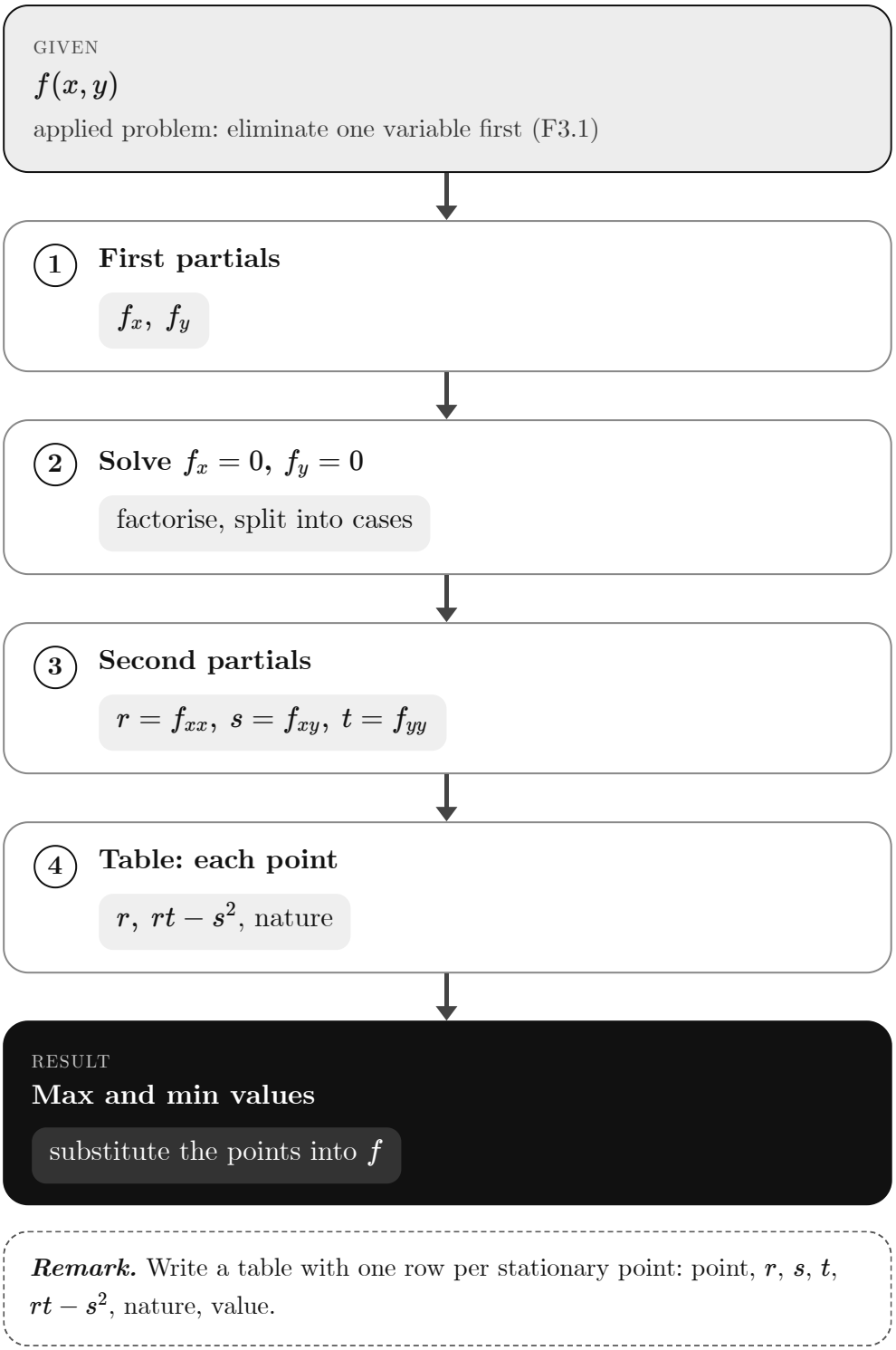
Class 010 · Problems on maxima and minima for a function of two variables · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S010·8	Deck 010, slide 8 (rendered .pptx)
T010a 00:07:38	AV-summary transcript “Extremas of functions continued” (older lecture), segment start
T010b 00:03:12	AV-summary transcript “Problems on Maxima and Minima for a function of two variables” (newer lecture)
CW U1 P29	Unit 1 class-work problem 29
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c
ESA Jan-24 s2 1a	Past paper: Jan 2024 set 2, question 1.a

Contents

1. Figure F1.1 · Solving a maxima–minima problem
2. Figure F1.2 · Classifying each stationary point
3. Figure F3.1 · Applied problem by elimination
4. Stationary points of the two ESA functions

1 The method, and the ESA cubic functions



Remark. Write a table with one row per stationary point: point, r, s, t, $rt - s^2$, nature, value.

Figure F1.1. Solving a maxima–minima problem ESA ★★★

S010·7 · S010·8 · T010b 00:05:00

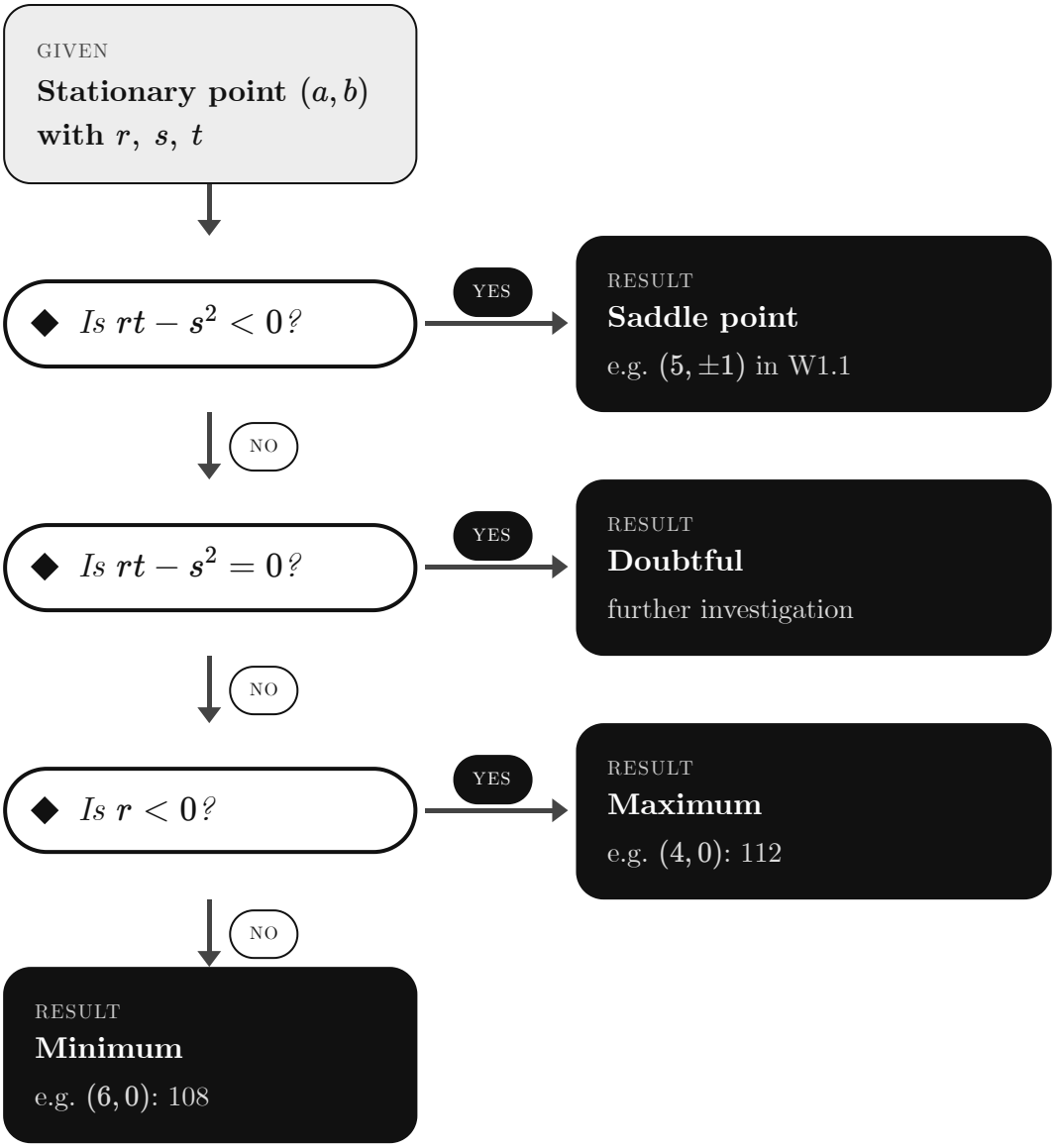


Figure F1.2. Classifying each stationary point ESA ★★★

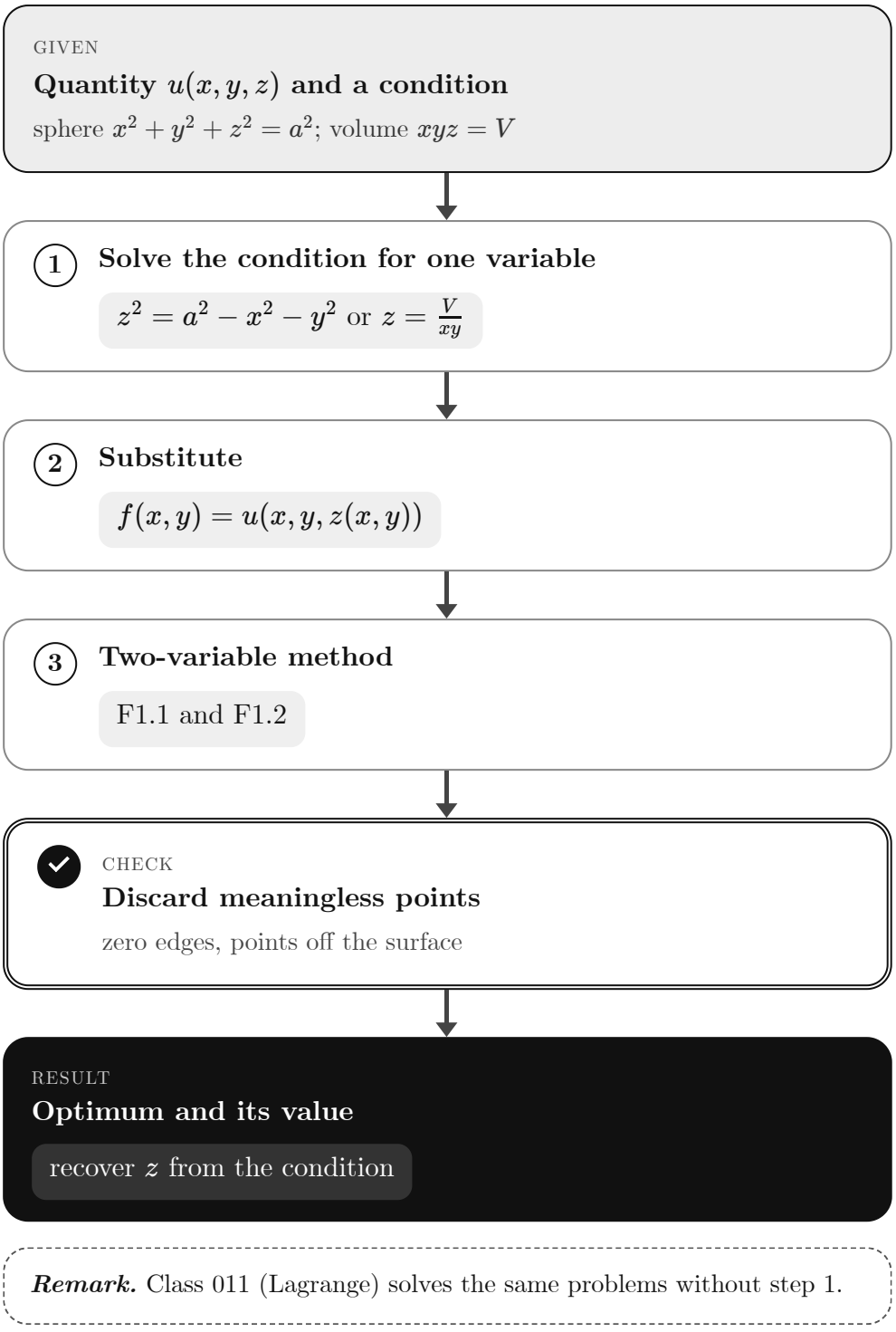
S010·8 · T010b 00:04:02

Stationary points of the two ESA functions

Function	Point	r	$rt - s^2$	Nature	f
$x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$	$(4, 0)$	-6	36	Maximum	112
	$(6, 0)$	6	36	Minimum	108
	$(5, \pm 1)$	0	-36	Saddle	—
$x^3 - 12x + y^3 + 3y^2 - 9y$	$(-2, -3)$	-12	144	Maximum	43
	$(2, 1)$	12	144	Minimum	-21
	$(2, -3)$, $(-2, 1)$	± 12	-144	Saddle	—

S010·8 · ESA Dec-23 1c · ESA Jan-24 s2 1a

3 Applied problems: eliminate one variable first



Remark. Class 011 (Lagrange) solves the same problems without step 1.

Figure F3.1. Applied problem by elimination ESA ★★

S010·3 · S010·5 · T010a 00:00:40 · T010a 00:08:27